Case Study on Deluge Scripting: Writing Custom Functions and Basic Programs

Name:	Irwin Inbadurai V.J	**Reg. No.:**	241201018
Course:		**Date:**	

Aim

To understand Deluge, the scripting language of Zoho Creator, and to write basic scripts and a custom function that add business logic to a fee management application.

Objectives

- Learn the purpose and structure of Deluge.
- Identify the form events in which scripts can run.
- Write validation, record-fetching and e-mail scripts.
- Create and call a custom function.

About Deluge

Deluge stands for Data Enriched Language for the Universal Grid Environment. It is the online scripting language integrated with Zoho Creator and is used to modify forms dynamically, execute tasks such as sending mail or inserting records, and define functions that can be called from other scripts. Its syntax is English-like, and scripts can be assembled with a drag-and-drop builder as well as typed. Scripts run on the server, and Zoho publishes limits on how much a single function may execute, so loops must be written carefully.

Selected Application

Tuition Centre Fee Manager (illustrative). Students pay monthly fees. The centre wants the application to reject invalid amounts, calculate late fees automatically, and e-mail a receipt after every payment. These needs cannot be met by form fields alone and therefore require Deluge.

Where Scripts Run

Event	When it executes	Typical use
On Load	When form opens to add or edit	Set default date, hide a field
On Validate	Before data is saved	Check ranges, block duplicates
On Success	After data is saved	Send mail, insert records
On User Input	When a field value changes	Fill dependent fields
Schedule	At set time or date field	Monthly reminders
Approval	On approve or deny	Update other forms

Basic Building Blocks

Concept	Example	Meaning
Variable	fee = 1500;	Stores a value
Input field	input.Amount	Value typed in the form
Condition	if (x > 0) { ... }	Decision making
Collection	list, map	Group of values
Loop	for each r in records	Repeat for every item
Debug	info fee;	Shows value in log to developers

Program 1: Validation (On Validate)

```
if (input.Amount_Paid <= 0)
{
    alert "Amount must be greater than zero";
    cancel submit;
}
```

The script stops the submission and shows a message when the amount is zero or negative.

Program 2: Fetch Record and Fill Fields (On User Input)

```
stu = Student[Register_No == input.Register_No];
input.Student_Name = stu.Name;
input.Monthly_Fee  = stu.Fee;
```

When a register number is chosen, the matching student record is fetched and the name and fee fill in automatically.

Program 3: Custom Function for Late Fee

A function is written once and called wherever needed, which avoids repeating logic in several forms.

```
decimal calcLateFee(decimal fee, int daysLate)
{
    late = 0;
    if (daysLate > 0 && daysLate <= 7)
    {
        late = fee * 0.02;
    }
    else if (daysLate > 7)
    {
        late = fee * 0.05;
    }
    return late;
}
```

Calling the function in the form script: `input.Late_Fee = thisapp.calcLateFee(input.Monthly_Fee, lateDays);` (the exact call prefix depends on how the function is created in the application).

Program 4: Receipt E-mail (On Success)

```
sendmail
[
    from: zoho.adminuserid
    to: input.Parent_Email
    subject: "Fee receipt"
    message: "Payment of Rs. " + input.Amount_Paid + " received. Thank you."
]
```

Late Fee Logic Check

Fee (Rs.)	Days late	Rule applied	Late fee (Rs.)
1,500	0	None	0
1,500	5	2%	30
1,500	10	5%	75
2,000	10	5%	100

Good Practices

- Put each piece of logic in the event that actually fires when it is needed.
- Test with `info` statements, then remove them; use meaningful names and comments.
- Zoho's Zia can draft Deluge from a plain-language description; the result should always be read and tested before deployment.

Results

- Validation, auto-fill, calculation and mail automation took under 40 lines of script; the late-fee rule now lives in one function.

Conclusion

Deluge extends a no-code application with exactly the logic that drag-and-drop settings cannot provide. Its readable syntax makes it suitable for beginners, and events and functions keep scripts organised. Learning a few patterns, validate, fetch, calculate and notify, covers a large share of real business requirements.

Sources consulted: Zoho Creator Help documentation (forms, reports, pages, workflows, Deluge); Zoho Corporation press release on Creator AI features (April 2025) and its coverage in IT Brief; Zoho Creator Decode article on Zia-assisted Deluge (March 2026). The business scenario in this report is illustrative, and figures in it are assumed values used to demonstrate the method, not measured results.